MULTIPLEXORES

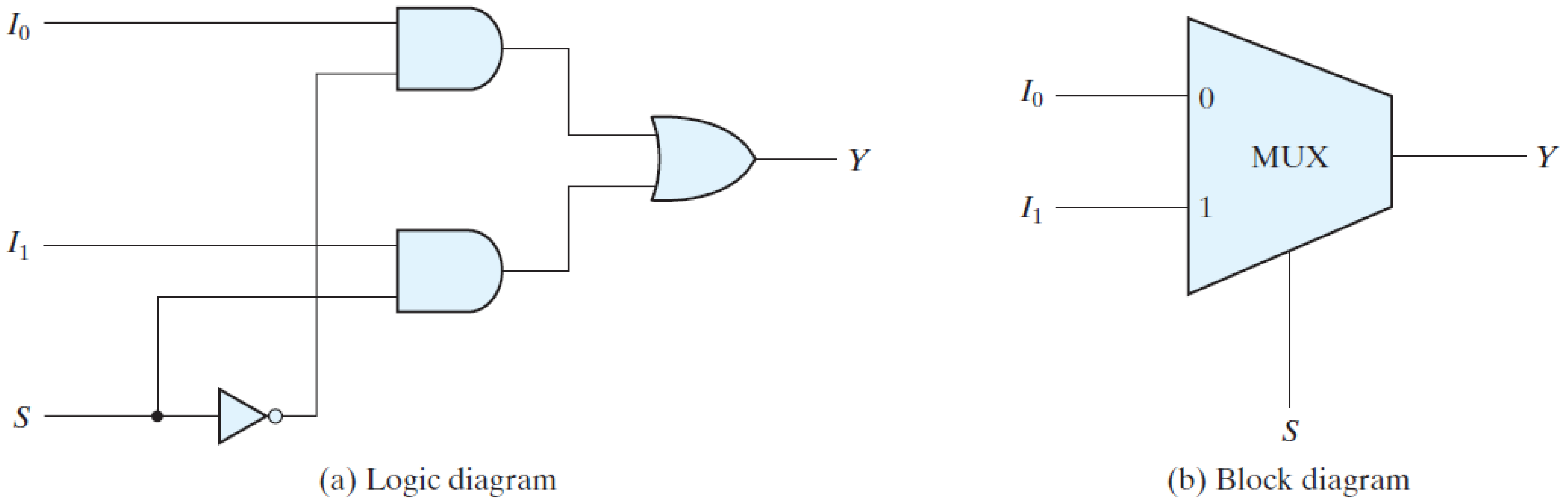
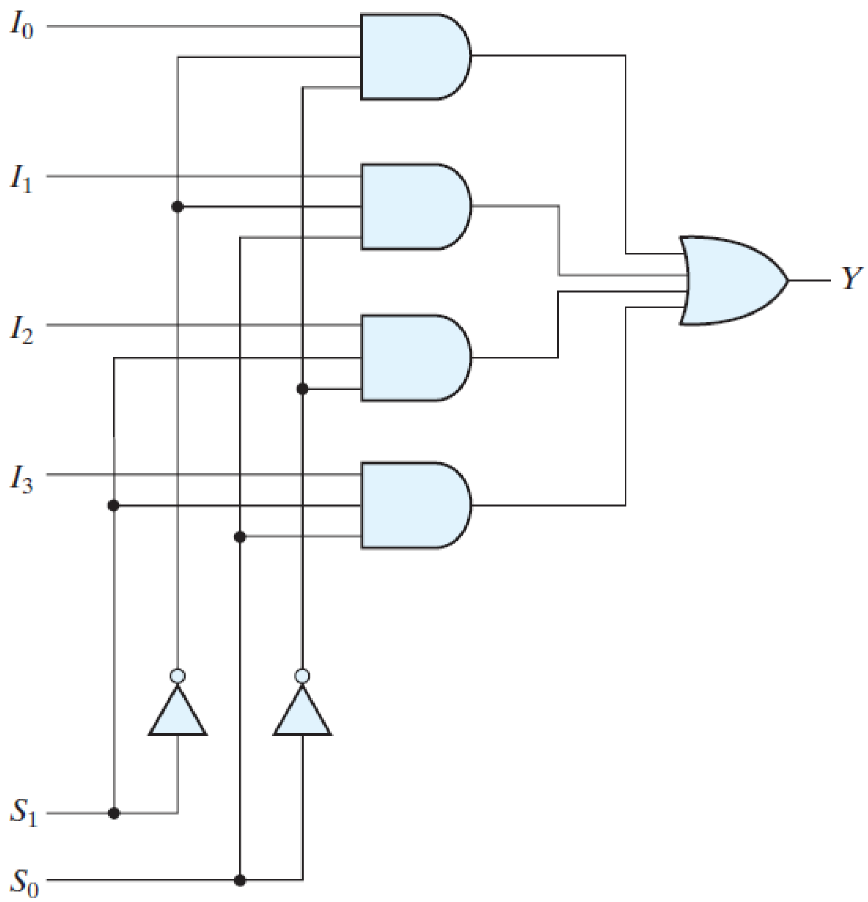


FIGURE 4.24
Two-to-one-line multiplexer

MULTIPLEXORES



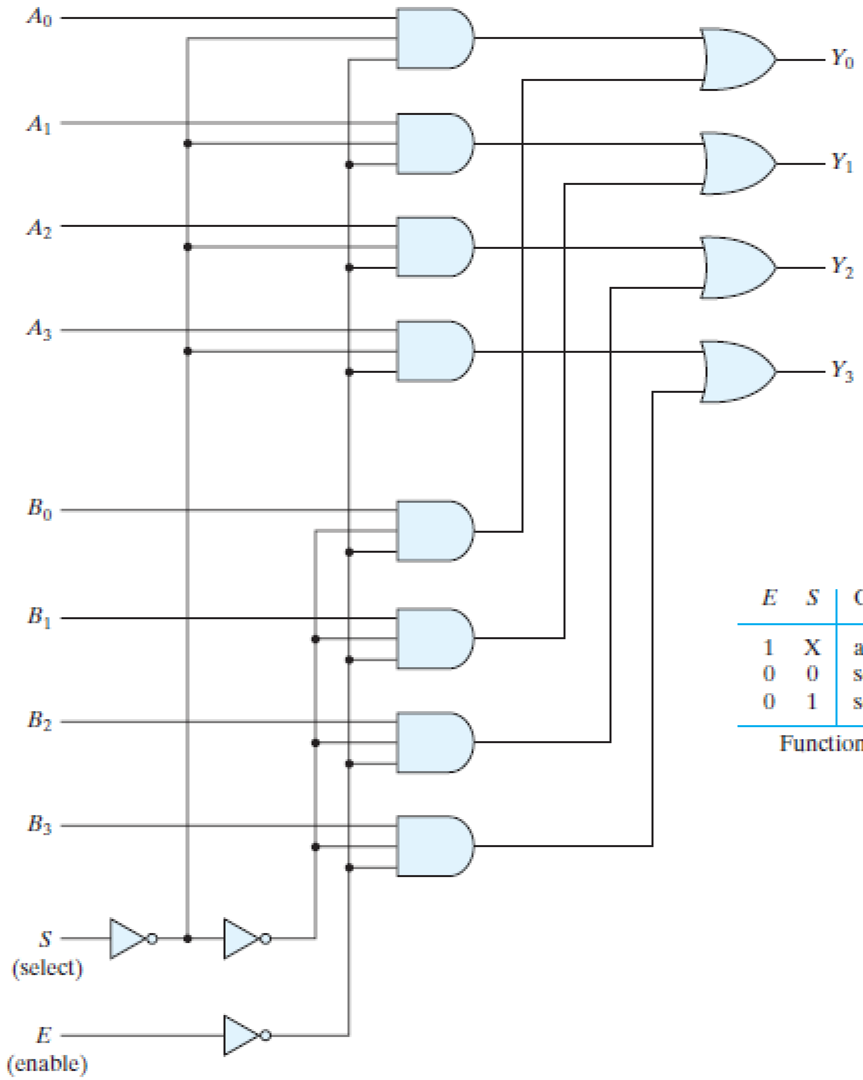
(a) Logic diagram

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

(b) Function table

FIGURE 4.25
Four-to-one-line multiplexer

MULTIPLEXORES

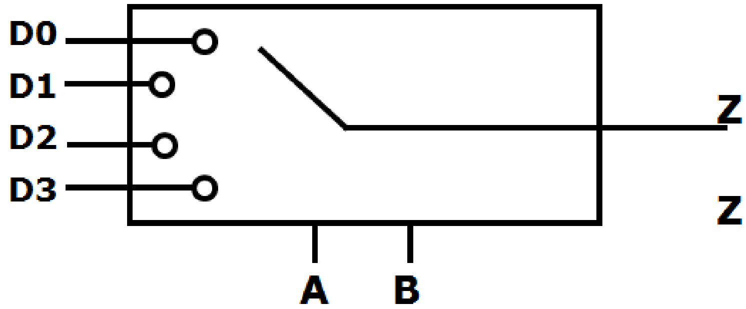


E	S	Output Y
1	X	all 0's
0	0	select A
0	1	select B

Function table

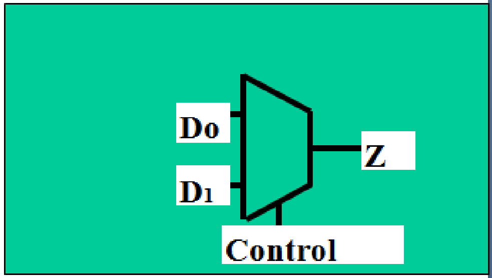
FIGURE 4.26
Quadruple two-to-one-line multiplexer

MULTIPLEXORES

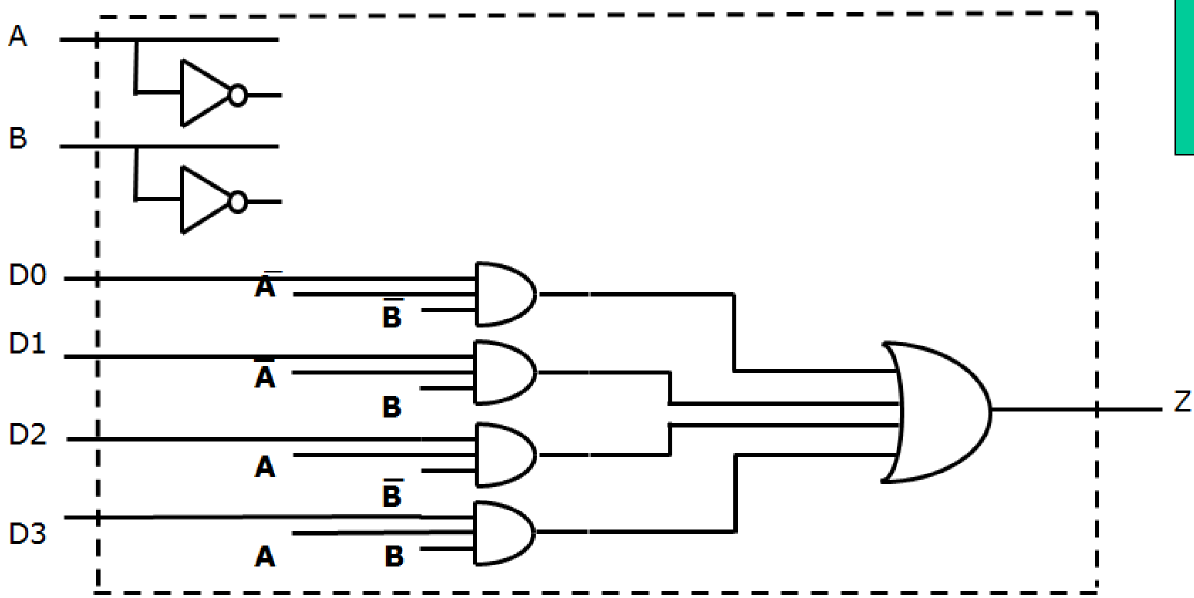


$$Z = D0\,\bar{A}\,\bar{B} + D1\,\bar{A}\,B + D2\,A\,\bar{B} + D3\,A\,B$$

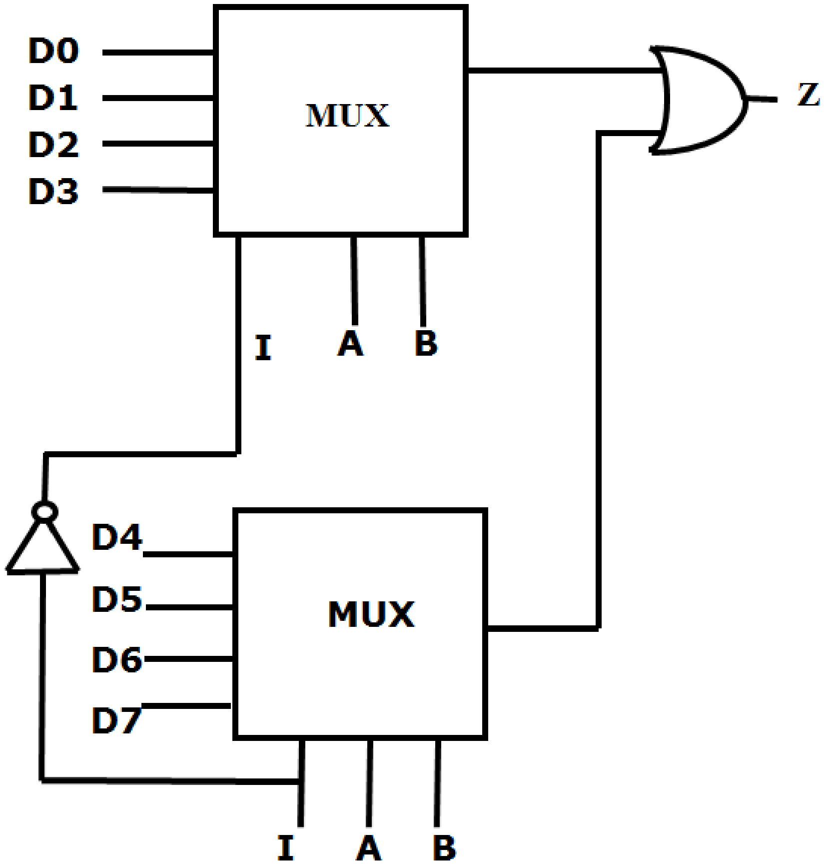
$$Z = D0\,m0 + D1\,m1 + D2\,m2 + D3\,m3$$



CONTROL		SALIDA
A	B	Z
0	0	D0
0	1	D1
1	0	D2
1	1	D3



MULTIPLEXORES



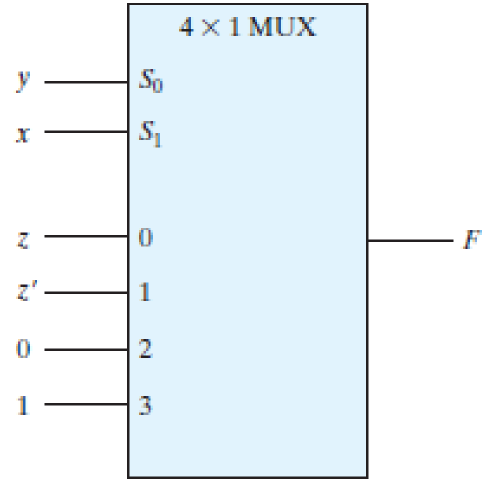
ENTRADAS			SAL.
I	A	B	Z
0	0	0	D0
0	0	1	D1
0	1	0	D2
0	1	1	D3
1	0	0	D4
1	0	1	D5
1	1	0	D6
1	1	1	D7

I : ENTRDA DE HABILITACION

MULTIPLEXORES

x	y	z	F	
0	0	0	0	$F = z$
0	0	1	1	
0	1	0	1	$F = z'$
0	1	1	0	
1	0	0	0	$F = 0$
1	0	1	0	
1	1	0	1	$F = 1$
1	1	1	1	

(a) Truth table



(b) Multiplexer implementation

FIGURE 4.27
Implementing a Boolean function with a multiplexer

$$F(A, B, C, D) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$$

MULTIPLEXORES

A	B	C	D	F	
0	0	0	0	0	$F = D$
0	0	0	1	1	
0	0	1	0	0	$F = D$
0	0	1	1	1	
0	1	0	0	1	$F = D'$
0	1	0	1	0	
0	1	1	0	0	$F = 0$
0	1	1	1	0	
1	0	0	0	0	$F = 0$
1	0	0	1	0	
1	0	1	0	0	$F = D$
1	0	1	1	1	
1	1	0	0	1	$F = 1$
1	1	0	1	1	
1	1	1	0	1	$F = 1$
1	1	1	1	1	

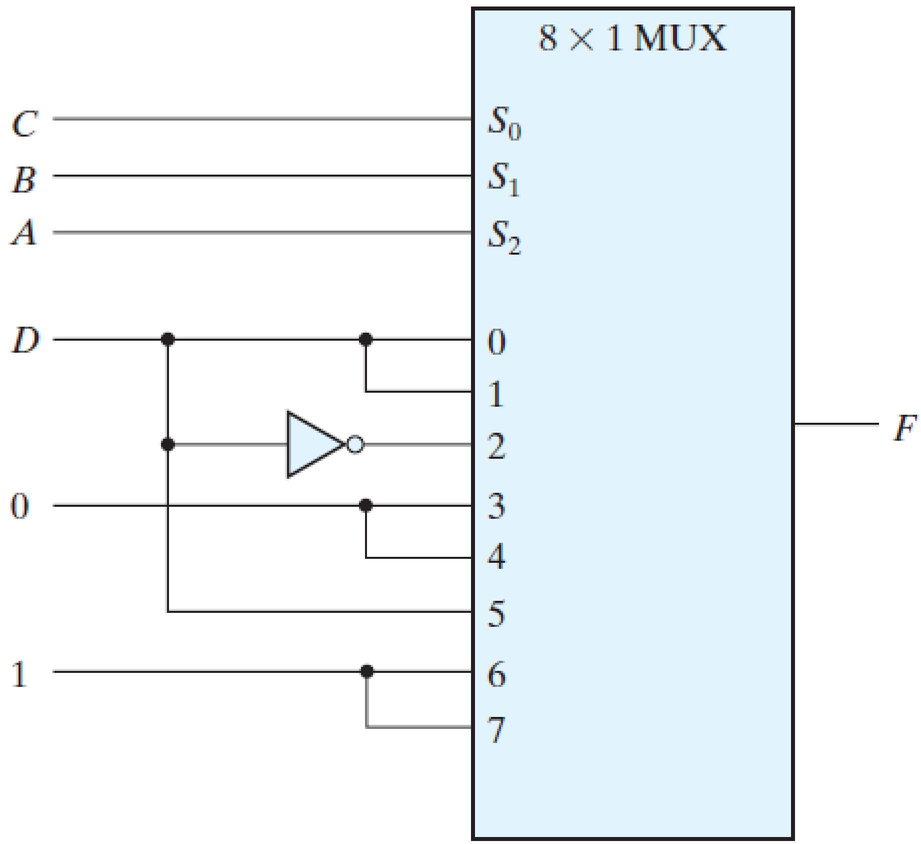
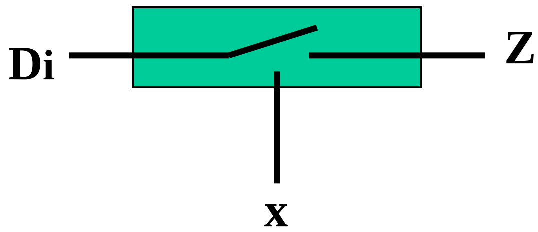
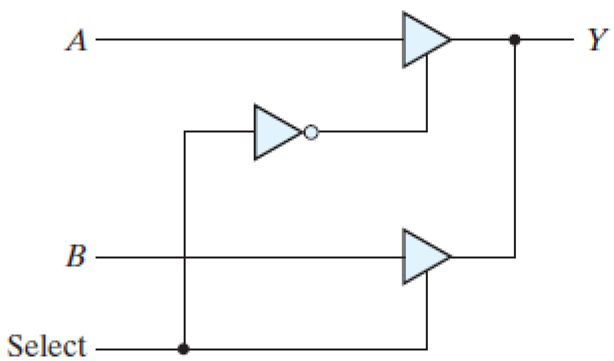
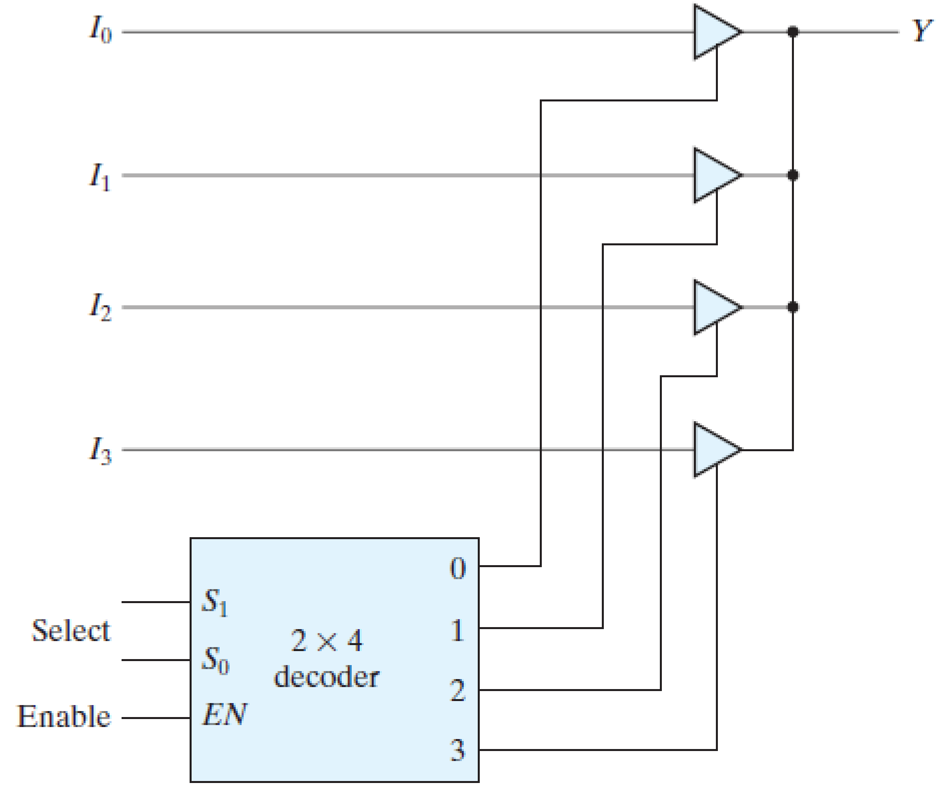


FIGURE 4.28
Implementing a four-input function with a multiplexer

COMPUERTAS CON TERCER ESTADO







(a) 2-to-1-line mux (b) 4-to-1-line mux

FIGURE 4.30
Multiplexers with three-state gates